

SUKHMANJOT AULAKH

☎ +1(647) 907-3037 📍 Toronto, ON

✉ sukhmanjot.aulakh@torontomu.ca 🌐 sukhmanjot-aulakh 📺 SukhmanAulakh

EDUCATION

Bachelor of Engineering - Computer Engineering (Software)

Expected 2027

Toronto Metropolitan University

Toronto, ON

Operating Systems, Algorithms, Object-Oriented Programming, Database Systems, Compilers, Software Testing & QA.

Certificate of Completion

Jun 2025

The Joy of React, Josh Comeau

Online

React Fundamentals, State Management, React Hooks, Component API Design, Prop Delegation.

TECHNICAL SKILLS

Programming Languages
Frameworks & Libraries
Cloud, DevOps & Tools
Databases & Concepts

C, C++, Python, Java, TypeScript, JavaScript, SQL, HTML/CSS
React.js, Next.js, FastAPI, Flask, Tailwind CSS, Bootstrap, Mapbox GL JS, HeroUI
AWS EC2, Docker, Kubernetes, Nginx, Git, GitHub Actions, Firebase, Heroku, GA4, Arduino
PostgreSQL, PostGIS, MySQL, REST APIs, CI/CD, SSG, SEO, Agile, Linux

EXPERIENCE

Electrical Lead

Jan 2025 – Present

Toronto Metropolitan Baja Racing

Toronto, ON

- Developed embedded software in **C++** and **Arduino** for custom sensors including RPM, gear position, and oil temperature, enabling real-time data collection and reverse light activation.
- Led a team of **15+ engineers** using Agile practices to design and implement the electrical system for an SAE off-road vehicle while ensuring compliance with SAE standards.

Web Developer

Apr 2024 – Present

Toronto Metropolitan Baja Racing

Toronto, ON

- Engineered a high-performance web application using **Next.js 15**, **React**, and **TypeScript**, establishing a professional digital presence for the university's oldest engineering design team.
- Architected an automated **CI/CD** pipeline with **GitHub Actions** for seamless deployment to **Firebase Hosting**, eliminating manual deployment overhead.
- Integrated **Google Analytics 4** and **Google Drive APIs** to track user engagement and enable non-technical leads to update gallery and sponsor content without code changes.
- Designed a racing-focused UI system with **Tailwind CSS** and **HeroUI**, optimizing performance and SEO through **SSG** for near-instant load times.

PROJECTS

Ontario Crown Land Terrain Finder

Mar 2026

- Developed a full-stack GIS application using **React.js**, **Next.js**, **FastAPI**, and **Python**, using **Mapbox GL JS** for 3D terrain visualization and path measurement.
- Built high-performance **REST APIs** with **PostgreSQL** and **PostGIS** to process geospatial data and stream vector tiles for map rendering.
- Deployed a containerized production stack on **AWS EC2** using **Docker**, **Nginx**, and **Let's Encrypt HTTPS** for secure hosting.

Farm2Cart

May 2025

- Built a containerized **Java/Tomcat microservices** marketplace and deployed it using **Docker** and **Kubernetes**.
- Designed a layered architecture with a **JSP/Tomcat** frontend backed by dedicated **MySQL** databases.
- Implemented **REST APIs** and **token-based authentication** for secure backend integration.